2	**(a)**	**(i)**	horizontal component (= $12 \cos 50°$) = $7.7\,\mathrm{m\,s^{-1}}$	A1	[1]
		(ii)	vertical component (= $12 \sin 50°$ or $7.7 \tan 50°$) = $9.2\,\mathrm{m\,s^{-1}}$	A1	[1]
	(b)		$v^2 = u^2 + 2as$ <u>and</u> $v = 0$ or $mgh = ½mv^2$ or $s = v^2 \sin^2\theta / 2g$	C1	
			$9.2^2 = 2 \times 9.81 \times h$ hence $h = 4.3$ (4.31) m	A1	[2]
			alternative methods using time to maximum height of 0.94 s:		
			$s = ut + ½at^2$ and $t = 0.94$ (s)	(C1)	
			$s = 9.2 \times 0.94 - ½ \times 9.81 \times 0.94^2$ hence $s = 4.3$ m	(A1)	
			or		
			$s = vt - ½at^2$ and $t = 0.94$ (s)	(C1)	
			$s = ½ \times 9.81 \times 0.94^2$ hence $s = 4.3$ m	(A1)	
			or		
			$s = ½(u + v)t$ and $t = 0.94$ (s)	(C1)	
			$s = ½ \times 9.2 \times 0.94$ hence $s = 4.3$ m	(A1)	
	(c)		t (= 9.2 / 9.81) = 0.94 (0.938) s	C1	
			horizontal distance = 0.938×7.7 (= 7.23 m)	C1	
			displacement = $[4.3^2 + 7.23^2]^{1/2}$	C1	
			= 8.4 m	A1	[4]